Bridging Human Cognition and AI: Enhancing Transparency and Explainability with Hierarchical Conceptual Graphs and the Knowing Protocol

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Abstract

The rapid deployment of large language models (LLMs) in critical domains demands greater transparency and explainability to build user trust and enable effective collaboration. Current AI-human interactions largely rely on unstructured text, often resulting in misunderstandings and limited insight into AI reasoning. We introduce a Hierarchical Conceptual Graph Model and the Knowing Communication Protocol to bridge the gap between symbolic human reasoning and sub-symbolic AI processing. Our model combines conceptual spaces, ontologies, and hierarchical structures to explicitly represent complex knowledge, while the Knowing Protocol, through the Knowing Markup Language (KML), facilitates structured, machine-readable interactions. This approach enhances transparency by aligning AI-generated content with human cognitive structures, promoting clarity and collaborative knowledge building—ultimately addressing the limitations of traditional text-based AI tools and advancing trustworthy, explainable AI.

Keywords: Explainability; Transparency; Knowledge Representation; Hierarchical Conceptual Graphs; Knowing Communication Protocol; Human-AI Interaction; Large Language Models; Cognitive Alignment; Structured Communication; Collaborative Knowledge Building.

1. Introduction

Despite the remarkable capabilities of large language models (LLMs) such as GPT-3 and GPT-4, a significant gap remains in effective communication between humans and AI systems (Bender et al., 2021). Humans primarily rely on *symbolic representations* for reasoning and knowledge exchange, allowing for clear expression and manipulation of complex ideas (Gärdenfors, 2000). In contrast, LLMs utilize *sub-symbolic representations*, encoding information in high-dimensional spaces that are not readily interpretable by humans.

This fundamental difference creates a barrier to transparency and explainability in AI systems. Current AI-human interactions rely predominantly on unstructured text exchanges through prompt-response paradigms. This mode of communication places humans on less favorable ground—essentially "playing on the machine's ballpark"—where text serves as the main mediator. This setup favors AI processing over human cognitive structures. Tools like ChatGPT and Microsoft Copilot, while helpful, still lean heavily on text, which may not fully support the exchange of complex ideas and can often lead to misunderstandings, ambiguities, and inefficiencies (Andreas et al., 2018).

Without a bridge between symbolic human reasoning and sub-symbolic AI processing, users lack insight into the AI's reasoning processes embedded in it's output. This opacity hinders trust and collaboration, particularly in high-stakes domains where understanding the rationale behind AI decisions is crucial (Ribeiro et al., 2016). There is a pressing need for a communication framework that aligns AI outputs with human cognitive structures, enhancing transparency and enabling collaborative knowledge building.

In this paper, we propose a solution to bridge this gap:

1. We introduce a **Hierarchical Conceptual Graph Model**, a novel graph-based knowledge representation that integrates conceptual spaces, ontologies, and hierarchical structures to represent complex knowledge explicitly.

2. We present the **Knowing Communication Protocol**, which leverages the Knowing Markup Language (KML) to enable structured interactions between an application's hierarchical graph view and an LLM. This protocol facilitates knowledge exchange in a machine-interpretable format that aligns AI communication with human knowledge construction, reducing ambiguity and enhancing understanding.

Our approach enhances transparency and explainability in high-stakes environments by aligning knowledge representation with human cognitive structures and providing a structured communication protocol. It allows users to understand and interact with AI reasoning processes more effectively, fostering trust and enabling collaborative knowledge building.

2. Related Work

Knowledge representation is fundamental in artificial intelligence, enabling systems to model and reason about the world. Traditional models like semantic networks and ontologies represent hierarchical relationships and support inferencing (Sowa, 1984). Ontologies provide structured frameworks for representing domain knowledge with classes, properties, and relationships (Gruber, 1993). Conceptual graphs, introduced by Sowa (1984), offer graphical representations that combine the expressiveness of semantic networks with the rigor of formal logic but can become complex when representing multidimensional concepts.

Peter Gärdenfors proposed the theory of *Conceptual Spaces*, representing concepts as points or regions in a geometric space defined by quality dimensions (Gärdenfors, 2000). This model bridges symbolic and sub-symbolic representations, aligning closely with human cognitive processes. However, implementing conceptual spaces can be challenging due to high dimensionality and lack of standardized methods for defining dimensions across domains.

Explainability has become a critical concern in AI, especially with opaque models like deep neural networks. LLMs like GPT-3 and GPT-4 exacerbate explainability challenges due to their sub-symbolic nature and difficulty in interpreting their internal states (Bender et al., 2021). Users interact with these models primarily through unstructured text, leading to misunderstandings and a lack of transparency in the AI's reasoning process. Traditional knowledge representation models, while expressive, often struggle with scalability and explicit representation of multidimensional properties, making them less intuitive for users (Sowa, 1984; Gärdenfors, 2000).

Our proposed Hierarchical Conceptual Graph Model builds upon these foundations by integrating their strengths and addressing their limitations. By organizing knowledge into

Multidimensional Spaces, *Single-Dimensional Spaces*, and *Ontology Spaces*, our model explicitly represents complex concepts and their properties in a hierarchical and multidimensional framework. The Knowing Communication Protocol bridges symbolic human reasoning and sub-symbolic AI by enabling an LLM to interpret and interact with the structured Graph Model.

3. Proposed Approach

3.1. Hierarchical Conceptual Graph Model

Our *Hierarchical Conceptual Graph Model* integrates conceptual spaces, ontologies, and hierarchical structures into a directed acyclic graph (DAG) consisting of nodes and edges with specific roles and relationships.

Nodes are categorized into Spaces and Concepts, with adjustable granularity allowing users to decide how much sub-symbolic information is represented within each node. This flexibility supports scalability by enabling users to tailor the model's detail to the level of complexity needed, addressing a common challenge in knowledge graphs. Spaces define domains within the graph and are of three types: Multidimensional Space (MS), representing domains where multidimensional concepts reside; Single-Dimensional Space (SS), corresponding to single properties or qualities (e.g., color, size); and Ontology Space (OS), representing ontological structures organized by "IsA" relationships.

Edges define relationships: ParentChild edges represent hierarchical relationships within a Space; IsA edges represent inheritance relationships within OS and between MC nodes and OS concepts; Have edges connect MC nodes to SC nodes, indicating that a Multidimensional Concept has a specific property value from a Single-Dimensional Space.

Our model builds upon Gärdenfors' conceptual spaces, where concepts are represented in a geometric space with dimensions corresponding to qualities or properties (Gärdenfors, 2000). Single-Dimensional Spaces correspond to these dimensions, and Multidimensional Concepts are points defined by their positions along these dimensions. We also draw inspiration from ontologies and semantic networks, which use hierarchical structures and "IsA" relationships to represent knowledge (Sowa, 1984). Furthermore, our approach parallels Fahlman's Scone system, which employs multiple contexts to emulate human-like reasoning (Fahlman, 2006).

3.2. Knowing Communication Protocol

The *Knowing Communication Protocol* facilitates structured interactions between humans and LLMs using the *Knowing Markup Language (KML)*. KML is a minimalistic, human-readable, and AI-friendly markup language designed to represent hierarchical tree structures and track node states within different Spaces (Knowing Project, 2023).

When applied in an application displaying hierarchical data, this protocol allows users and AI to communicate through clear, structured formats rather than relying on ambiguous text exchanges. It enables precise communication of complex ideas and facilitates seamless integration with AI models while remaining accessible and understandable to human users.

The Knowing® application[1] implements this protocol using a fine-tuned LLM, creating a practical environment for structured interaction through KML.

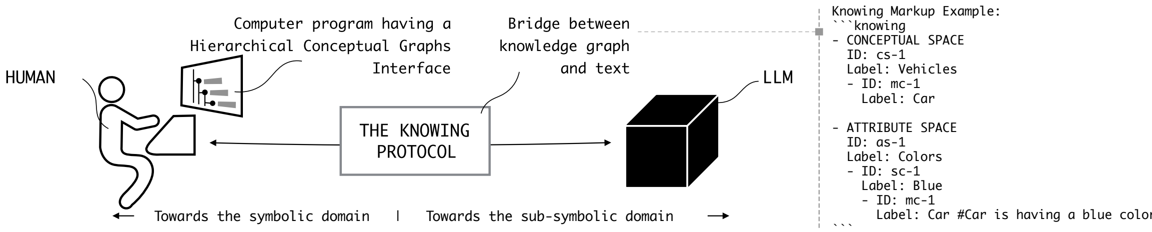

Figure 1: Enhanced transparency and explainability arise when the conceptualization process shifts from the implicit, sub-symbolic domain toward a more explicit, symbolic structure.

4. Enhancing Transparency and Explainability

Our Hierarchical Conceptual Graph Model and Knowing Communication Protocol together provide a clear and hierarchical structure for knowledge representation, enhancing transparency in AI systems. By organizing information into distinct Spaces—Conceptual, Attribute, and Ontology—the model enables users to see explicitly how concepts and their properties relate, making AI reasoning processes more understandable and interpretable (Gärdenfors, 2000).

The structured format of KML reduces ambiguity compared to traditional text-based exchanges. Unlike conventional text responses, where users may passively copy information without achieving genuine understanding, our approach encourages interaction with structured knowledge. The Knowing Protocol facilitate precise communication between a graph view application and an LLM, allowing users to build, explore, and refine a cohesive knowledge structure, minimizing miscommunication and aligning closely with natural human categorization and relational understanding (Norman, 2013).

5. Conclusion

We introduced a novel *Hierarchical Conceptual Graph Model* and the *Knowing Communication Protocol* to improve transparency and explainability in AI interactions. By combining conceptual spaces, ontologies, and hierarchical structures, our model provides an explicit and intuitive way to represent complex knowledge, while the Knowing Protocol enables structured interactions between an application and an LLM through KML, allowing users to view and modify information within a clear, hierarchical framework.

Our approach fosters trustworthy AI by aligning knowledge representation with human cognitive processes, reducing risks tied to opaque decision-making and allowing users to engage actively in knowledge building (Dignum, 2019).

1. https://docs.knowing.app/knowing_markup

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